AB-INITIO CALCULATIONS OF STRUCTURAL ELECTRONIC AND TRANSPORT PROPERTIES OF 2D GOLD MONOLAYER

A Dissertation

Submitted to the Dean office, Institute of Science and Technology Tribhuvan University, Kritipur in the Partial Fulfilment for the Requirement of Master's Degree of Science in Physics

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RECOMMENDATION

It is certified that Mr. Shambhu Bhandari Sharma has carried out the dissertation work entitled **'ab-initio calculations of structural electronic and transport properties of 2d gold monolayer'** under my supervision and guidance.

I recommend the dissertation in the partial fulfillment for the requirement of Masters' Degree of Science in Physics.

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EVALUATION

We certify that we have read this dissertation and in our opinion, it is satisfactory in the scope and quality as dissertation in partial fulfillment for the requirement of Masters' Degree of Science in Physics.

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Abstract

We performed first principle calculation based on Density Functional Theory (DFT) and Non-equilibrium Green Function (NEGF) approach within Generalized Gradient Approximation (GGA). We predict a highly stable and robust atomically thin gold monolayer with a rectangular lattice stabilized by metallic bonding. On reducing the dimension from 3D to 2D, we observed that the structural, electronic and transport properties changes drastically. The Au monolayer has a cohesive energy of 2.98 eV per atom but the Au bulk has a cohesive energy of 3.79 eV per atom respectively. Although the cohesive energy of the 2D Au monolayer is a smaller than that of the 3D bulk fcc phase,we found that monolayer has lower number of bonds per atom indicating that each bond in monolayer are stronger than in the bulk structure. The electronic band structure displays pure metallic nature of gold with increased number of bands crossing the Fermi level. In particular we studied the transport properties of gold monolayer considering electrode-device-electrode regime. We studied the transport properties with DFT-NEGF approach using the simulation module TranSIESTA. The transport properties is studied by eliminating the current quantization effect due to the contacts. By varying the the chemical potential in the electrode region at the step 0.1 V we obtained the current -voltage characteristics of the device and also obtained the transmission coefficient. The obtained plots of PDOS, DOS and transmission spectra of device region is varied with different bias voltage. The Current-voltage plot displays a negative differential resistance (NDR) mechanism which is comparable to the Resonant Tunneling Diode(RTD) with the peak-to-valley current ratio (PVCR) around 1.10.

List of Abbreviations

SIESTA	:	Spanish Initiative for Electronic Simulations with Thousands of Atoms
BO	:	Born-Oppenheimer
BZ	:	Brillouian Zone
TB	:	Tight Binding
NWs	:	Nanowires
CBM	:	Conduction Band Maximum
VBM	:	Valance Band Minimum
CP	:	Car Parrinello
CPMD	:	Car Parinello Molecular Dynamics
DFT	:	Density Functional Theory
DOS	:	Density of States
FPMD	:	First-Principles Molecular Dynamics
GGA	:	Generalized Gradient Approximation
HK	:	Hohenberg Kohn
KS	:	Kohn Sham
LDA	:	Local Density Approximation
PBE	:	Perdue, Burke, and Ernzerhof
PDOS	:	Projected DOS
PWscf	:	Plane Wave scf
RRKJ	:	Rappe-Rabe-Kaxiras-Joanopulas
SCF	:	Self Consistent Field
SWE	:	Schrodinger Wave Equation
TF	:	Thomas Fermi
USPP	:	Ultra Soft PseudoPotential
XC	:	eXchange Correlation

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Chapter 1

Introduction and Motivation

1.1 General Consideration

We use so many electronic devices to make our daily life comfort : mobiles, computers, TVs,laptops,sensing devices etc in our daily life.Within last few decades; the thickness and size of these microelectronic devices has been reducing in exponential ratio. The thickness of the transistors-like devices is approaching to 10th of nanometer to the atomistic limit. It is the age of nanotechonology where the nanodevices are abundant in number. Millions of nonomaterials have been designed,moduled and fabricated. The various properties of such a nanoparticle can not be explained by the classical picture as the explanation ultimately fails to explain the quantum mechanical behavior of nanomaterial in atomistic level.Smaller the object quantum mechanical effect are very very essential and relevant. As if a device is tenth of micron thick , distortion of some atoms from the device does not make so significant difference in device performance but in nanometer scale few no of atom change also make huge difference in output characteristics. In nano-molecular physics , not only the reducing the size of devices to the atomistic level (bottom-down)is done but also new approach so-called bottom up [1]design(buildng devices into the actual bigger devices from nano scale) has been implemented. In nanometer scale , its always not very easy to perform the meticulous work of fabrication and device performance testing. And within the last two decades have witnessed that computational virtual experiments have been so useful to model and experiment the devices before the fabrication. There is enormous advantages in research and development activities at the atomic and molecular levels by use of high performance computation. Computational materials design has opened up a new di-

mension in materials engineering and technology, providing exciting opportunities for new device ideas.

After the ground breaking discovery of graphene a 2d material, scientist have realized that two dimensional material have more promising properties.As we know, dimensionality is one of the most important parameters that influence material properties. The same chemical compounds can exhibit dramatically different properties depending on whether they are arranged in $0D, 1D, 2D$, or 3D [2] crystal structures. This is clearly demonstrated by the different carbon allotropes, including fullerenes [3] (0D) nanotubes [4] (1D) graphene (2D) [5]

As one of the noblest of all metals, unlike other transition metal catalysts, gold had long been viewed as catalytically inactive. While bulk gold is known as the most chemically inert metal in the periodic table, but drastically nanometer-sized gold particles are exceptionally active and most useful. We can consider the analogy of graphite(bulk) has huge diffrent then the two dimensional graphene and 1d nanotubes. Considering these fact ,Metal nanowires and in particular gold nanowires (NWs) have received a great deal of attention in past few years. Experiments on gold nanowires have prompted theoretical and simulation studies of the two dimensional monolayers of gold.

1.2 Nanotechnology

Nanotechnology is the study of the controlling the matter on an atomic and molecular scale. Generally nanotechnology deals with structures sized between 1-100 nanometers in at least one dimension, and involve modifying or developing materials within that size. It makes the material lighter, stronger, faster, smaller and more durable. Nanotechnology obligates the ability to frame components of molecular size and precise machine. In other words, nanotechnology refers to the contrived ability to construct items from the bottom up, using tools and techniques that are being defined to make high performance products [6]. According to National science Foundation, Nanotechnology is the capability to understand, manipulate and control matter at the level of individual atoms and molecules. Science and engineering are the primary operators of global technological competition. Modern science based on the unifying features of nature at the nano scale contributes a new foundation for innovation, knowledge,

and integration of technology. The past two decades have seen a bloom of research activities in nanoscience and a rapid development in delivering the potential of nanotechnology into diverse applications. The major driving force for these activities and development is that nanoscale materials can offer novel electronic, magnetic, optical, mechanical, and chemical properties, which lead to distinguished performance in many technological fields, such as mechanical engineering, electronics, information, and energy technologies [7].

1.3 Types Of Nanostructures

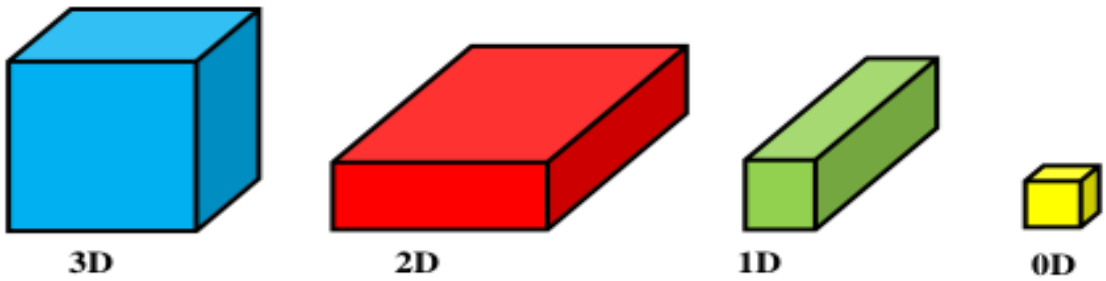

Figure 1.1: The Schematic representation of 3D, 2D, 1D and 0D systems [8].

Nanostructures are defined as the structures with at least one dimension less than 100 nm. In terms of dimensionality, nanostructures can exist in forms of zero-dimensional (quantum dot and clusters), one-dimensional (nanowires, nanotubes and nanoribbons), two-dimensional (atomic crystals and nanofilms) and 3D super structures made of nanoscale components. The schematic representation of 3D, 2D, 1D and 0D systems are shown in fig.1.1. These dimensions are in the sense of confinement of electron motion in the system e.g 3D system implies that the electrons are allowed to have random motion in all three directions, whereas 0D system implies complete confinement of electrons in all the three directions.

In particular, the 1D and 2D nanomaterials, termed as low-dimensional materials, are the ideal systems for studying the peculiar nanoscale effects, which are of benefit not only for understanding fundamental phenomena under these dimensionalities but also for developing functional nanodevices with high performance. Examples of these nanostructures are like Fullerene a Zero dimensional Structure, Carbon nanotubes are

the example of 1 Dimentional Structure, Graphene is a two dimensional structure and graphite is three dimentional struture are shown in fig.1.2

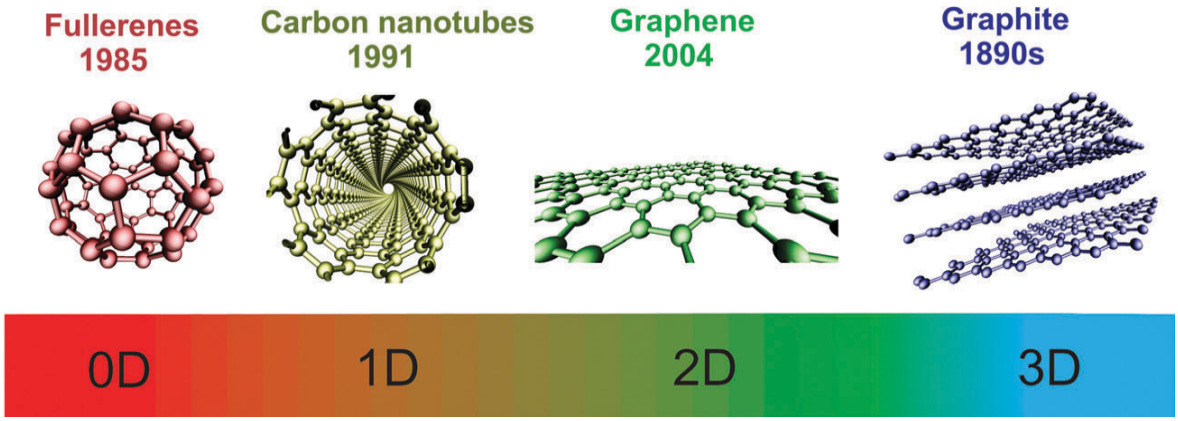

Figure 1.2: Examples of various dimensional nanostructures [9].

1.3.1 1D Nanostructures

In a 1D nanostructure, there is a confinement of electrons in two dimensions perpendicular to the longitudinal extent of the structure. This confinement is quantum mechanical in nature. It has been observed that 1D nanostructure represent the smallest dimension structure that can efficiently transport electrical carriers. Depending on topology and morphology, the 1D nanostructures can be classified into following way :

- Nanowires [10]
- Side-by-side biaxial nanowires [11]
- Nanoribbons [12]
- Nanotubes [13]
- Coaxial cable structures [14]

1.3.2 2D Nanostructures

Bulk crystals that are built up of van der Waals bonded layers can be separated by mechanical exfoliation into stable units of atomic thickness [16], which are also good candidates for growth by chemical vapour deposition and other techniques . These

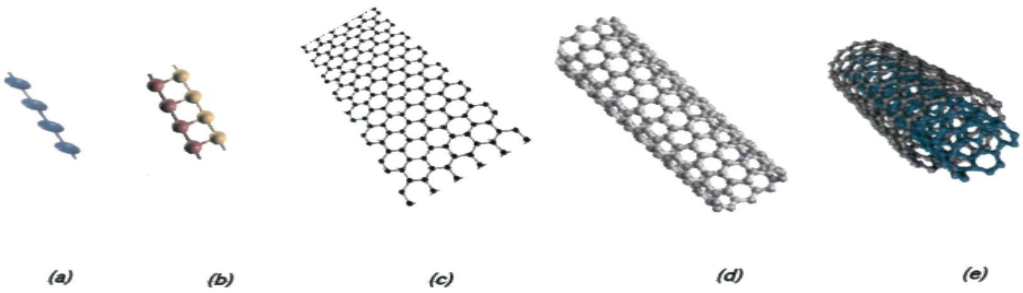

Figure 1.3: The schematic representation of various 1D structure (a) nanowire (b) side-by-side biaxial nanowire (c) Nanoribbon (d) Nanotube (e) coaxial cable nanostructure [15].

recently emerged 2D materials have demonstrated remarkable mechanical, electronic and optical properties that are absent in their bulk counterparts. These properties are widely tunable by doping, strain, external fields and environmental effects, owing to the materials atomic thickness]. With these unique properties, 2D atomic crystals have attracted tremendous scientific and technological interest. In the past few years, the discovery of graphene has intrigued vast studies on novel low-dimensional materials. Graphene is a two-dimensional material consisting of a single layer of carbon atoms arranged in a honeycomb structure. It is the basic structural element of other allotropes, including graphite, charcoal, carbon nanotubes and fullerenes. Its properties include high tensile strength and good conductivity of heat and electricity [20]. The stacked form of graphene is graphite. It is the material that has attracted the most interest due to its unique electronic properties. Graphene, a single-layer of graphite, two-dimensional sheet of hexagonally arranged carbon atoms. Finite termination of graphene results in quasi one dimensional nanoribbon like structures with two different possible edge geometries, namely zigzag and armchair, as shown in Fig. They are termed as zigzag graphene nanoribbons (ZGNRs) and armchair graphene nanoribbons (AGNRs), respectively and show very different electronic properties arising from their contrasting boundary conditions.

Gold has been playing a special role in human society through- out history. This unique role is closely related to its unusual resistance to oxidation, and also its exceptional stability during chemical reactions, extreme pressures, and elevated temperatures. [21] The exceptional stability of bulk gold is mainly ascribed to its strong relativistic effect and the so-called aurophilic attractions. As one of the noblest of all metals, unlike

other transition metal[22] catalysts, gold had long been viewed as catalytically inactive. While bulk gold is known as the most chemically inert metal in the periodic table, nanometer-sized gold particles are exceptionally active as catalysts in a wide range of chemical transformations. It has been shown that the catalytic properties of gold nanoparticles are sensitive to many factors, including the nature of the sub- strate, and the size, shape, and charge state of the gold particles.In an exciting recent development, free-standing atomically thick iron membranes suspended in graphene pores have been fabricated experimentally. These iron layers were small with just up to 10 atoms wide inside the pores. These iron patches had a different structure than the predicted full 2D isolated layer. This demonstrates the potential of perforated graphene as a support for small 2D membranes, and paves the way for novel 2D structures to be formed. And density functional theory has been very successful to describe the various properties of this type of system.

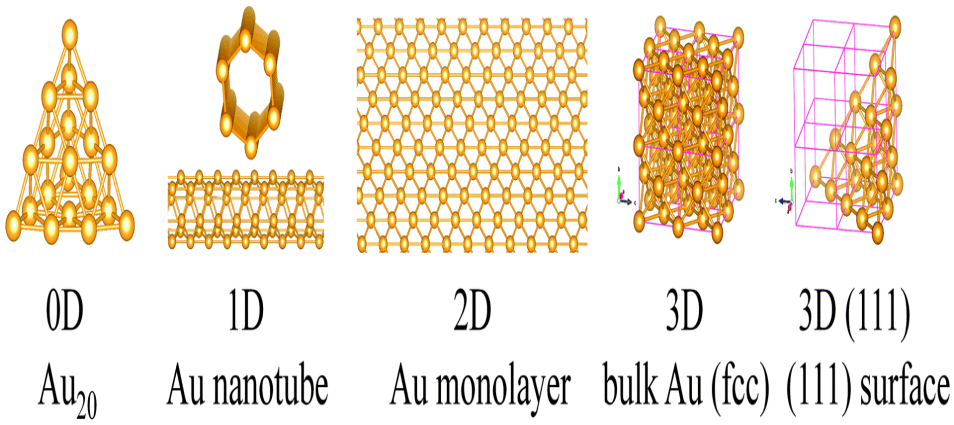

Figure 1.4: differernt dimensional structures of gold[23].

1.4 Review of literature

A lot of literature are available on the structural, electronic,mechanical properties of 1d,2d and 3d bulk strucutre of gold but the transport properties of gold monlayer hasnot been reported yet.

Zhao et al. have used density functional theory (DFT) calculations to study transitional metal systems, and predicted that the largest thermodynamically stable gold 2d monolayer can be fabricated.[23]

G. C. Bond et.al studied the catalytic properties of gold nanoparticles and reported that the catalytic properties of gold nanoparticles are sensitive to many factors, including the nature of the substrate, and the size, shape, and charge state of device.[24]

J. Zhao et.al fabricated the two dimensionally the possible structure of graphene.[25]

Kumar.A et.al reported the optical electronic and structural properties of ultra thin nanowire of gold and suggested that these one dimensional wires are transparent in visible region.[26]

M.Haun et .al noted that the Au(111) bulk surface can be used as a substrate for adsorption and surface science studies[28].

Melo Souza and et.al studied the Stretching of BDT-gold molecular junctions of thiol or thiolate termination and reported the transport properties of the structure. [29]

Yutaka.Hirose et.al studied the Nature of electron transport by pyridine-based tripodal anchors potential for robust and conductive single molecule junctions with gold electrodes[30]

Yan.et.al studied the Electron transport in Nano Gold-Silicon interfaces and confirmed the interaction of contacts in linear fashion[31]

1.5 Scope of Present Work

In this research we have modeled a two dimensional monolayer of gold and relaxed the geometrical structure to obtain its stable state by using the conjugate gradient approximation method. We have studied the structural electronic and transport properties of gold monolayer by using the DFT-NEGF approach. We obtained the density of states and band structure of gold monolayer and compared it with bulk structure. We obtained the total density of state and projected density of state and estimate the nature of solid i.e. metal, non metal, semiconductor or insulator can be defined by band structure. Magnetic properties of solid can be obtained by density of states(DOS). From DOS we can defined the whether solid is paramagnetic, ferromagnetic, diamagnetic or non magnetic. The main aim of our research is focused on studying the structural, and electronic properties of and designed model of two dimensional monolayer of gold.To find out the contribution of different orbitals in band structure and density of states

(DOS) we calculated projected partial density of states (PDOS). The Structural electronic and bulk calculations are performed under the DFT level of approximation, implemented with SIESTA and transport properties were performed under the NEGF approximation Using TranSIESTA

In chapter 1 , we have discussed the materials of our intrest and brief introduction of the reason of interest on it.

In chapter 2, we have discussed the theoretical basis of the method employed in our calculation. We have briefly discussed Born-Oppenheimer approximation, Hartree-Fock method and density functional theory.

In chapter 3, we discuss the computational details of our present work wih first principles pseudo-potential based density functional calculations by SIESTA package along with its execution process.

Chapter 4 contains the main findings of the present work along with the discussion.The determination of kinetic energy cutoff , lattice parameters and k-points were performed during convergence test calculation.Pseudopotential was generated with relativistic core correction was done by using ATOM program which we use in further calculations are discussed in this chapter. The band structure and the density of states of bulk Gold is thoroughly studied and analysed. Transport properties are calculated by considering electrode-device-electrode regime and compare the computated data with literatures which are in good agreement.

Chapter 5 gives the summary of research work and also inspires for the future work. The references used in the present study are listed at the end of thesis.

Chapter 2

Theoretical Background

2.0.1 The Many-Body System and Born-Oppenheimer (BO) Approximation

The Hamiltonian of a many-body condensed-matter system consisting of nuclei and electrons can be written as:

$$H_{tot} = -\sum_{I} \frac{\hbar^2}{2M_I} \nabla^2_{\mathbf{R}_I} - \sum_{i} \frac{\hbar^2}{2m_e} \nabla^2_{\mathbf{r}_i} + \frac{1}{2} \sum_{I,JI\neq J} \frac{Z_I Z_J e^2}{|\mathbf{R}_I - \mathbf{R}_J|} + \frac{1}{2} \sum_{i,j,i\neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{I,i} \frac{Z_I e^2}{|\mathbf{R}_I - \mathbf{r}_i|} \tag{2.1}$$

where, the indexes I, J run on nuclei, i and j on electrons, R_I and M_I are positions and masses of the nuclei, r_i and m_e of the electrons, Z_I the atomic number of nucleus I. The first term is the kinetic energy of the nuclei, the second term is the kinetic energy of the electrons, the third term is the potential energy of nucleus-nucleus Coulomb interaction, the fourth term represents the potential energy of electron-electron Coulomb interaction and the last term is the potential energy of nucleus-electron Coulomb interaction. The time-independent Schrodinger equation for the system reads:

$$H_{tot}\psi\{(\mathbf{R}_I), (\mathbf{r}_i)\} = E\psi\{(\mathbf{R}_I), (\mathbf{r}_i)\} \tag{2.2}$$

Where, $\psi\{(\mathbf{R}_I), (\mathbf{r}_i)\}$ is the total wave function of the system. If one can solve the above Schrondinger equation, everything about the system is known. However, it is impossible to solve it in practice, so Born-Oppenheimer approximation was made by Born and Oppenheimer in 1927. Since, the nuclei are much heavier than electrons (the mass of a proton is about 1836 times the mass of an electron), the nuclei move much slower (about two order of magnitude slower) than the electrons. Therefore, we can

treat the movement of nuclei and electrons separately. Then the total wave function can be written as,

$$\psi\{(\mathbf{R}_I),(\mathbf{r}_i)\} = \Theta\{(\mathbf{R}_I),\phi\{(\mathbf{r}_i);(\mathbf{R}_I)\}\} \tag{2.3}$$

Where, $\Theta\{(\mathbf{R}_I)\}$ and $\phi\{(\mathbf{r}_i);(\mathbf{R}_I)\}$ describes the nuclei and electrons respectively. By using the BO approximation above equation can be divided into two separate S.E.,

$$H_e\phi\{(\mathbf{r}_i),(\mathbf{R}_I)\} = V\{(\mathbf{R}_I)\}\phi\{(\mathbf{r}_i);(\mathbf{R}_I)\} \tag{2.4}$$

where,

$$H_e = -\sum_i \frac{\hbar^2}{2m_e}\nabla^2_{\mathbf{r}_i} + \frac{1}{2}\sum_{I,JI\neq J}\frac{Z_I Z_J e^2}{|\mathbf{R}_I-\mathbf{R}_J|} + \frac{1}{2}\sum_{i,j,i\neq j}\frac{e^2}{|\mathbf{r}_i-\mathbf{r}_j|} - \sum_{I,i}\frac{Z_I e^2}{|\mathbf{R}_I-\mathbf{r}_i|} \tag{2.5}$$

and

$$\left[-\sum_I \frac{\hbar^2}{2M_I}\nabla^2_{\mathbf{R}_I} + V\{\mathbf{R}_I\}\right]\Theta\{\mathbf{R}_I\} = E'\Theta\{\mathbf{R}_I\} \tag{2.6}$$

The eigen value energy $V\{\mathbf{R}_I\}$ depends parametrically on the positions of the nuclei. After, solving equation (2.4), $V\{\mathbf{R}_I\}$ is known and applying on equation (2.6), which has no electronic degree of freedom, the motion of the nuclei is obtained. The significance of BO approximation is to separate the movement of electrons and nuclei.

2.0.2 Hartree-Fock Approximation

Hartree-Fock (HF) approximation is also called self-consistent field or mean field. HF methods were introduced by Hartree and developed by Slater, Fock and others in the late 1920 to study the electronic structure of atoms with more than one electron. These HF methods are widely used to compute properties of atoms, molecules and solids. It is also used for the determination of the wave function and the energy of quantum many body system (i.e. N-body) in a ground state. The two basic approximation in this method are to treat the atomic nuclei are infinitely heavy and therefore fixed in space at their equilibrium positions, and to approximate the many electron wave function as a product of one electron wave functions that is fully antisymmetrized because electrons are fermions which must obey the Pauli exclusion principle. The difference of the exact non-relativistic ground state energy of the system and HF energy with the infinite basis set is called the correlation [33] or the error in HF approximation is known

as correlation. For large Z value , Hartree and exchange dominates over correlation but when Z is order of one, correlation becomes important.

The N electron anti-symmetric product of N single particle spin-orbitals is conveniently written as a determinant, known as the Slater determinant named after John Slater,

$$\psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(r_1, s_1) & ..... & \phi_N(r_1, s_1) \\ ..... & ..... & ..... \\ \phi_N(r_N, s_N) & ..... & \phi_N(r_N, s_N) \end{vmatrix}$$

If the two exchange particles are equivalent to the exchange of two columns, then the determinant sign has change. If two rows are equal determinant is zero, this demonstrate Pauli's exclusion principle, i.e. two or more identical fermions can not occupy the same state.

2.0.3 Density Functional Theory

Density functional theory (DFT) is the system in which find the ground state properties without dealing directly with the many electron state. Old DFT is Thomas-Fermi theory in 1926. Modern DFT begins with Kohn-Sham equations by introducing orbitals in and he also suggested Local Density Approximation (LDA) and Gradient Expansion Approximation (GEA) in 1965. Then in 1998 Walter Kohn was awarded with the Novel Prize in Chemistry for his development of DFT. It is presently most successful approach to complete the electronic structure of matter. Its applicability ranges from atoms, molecules and solids to nuclei, quantum and classical fluids. It is a computational quantum mechanical method used in Physics, Chemistry and material sciences. It is most popular and versatile methods available in condensed matter physics, computational physics and computational chemistry and computational cost is also low as compared to traditional methods i.e. Hatree-Fock method and other methods.

In DFT the complicated wave function ψ is replaced by electron density. The main idea of DFT is to describe a many-body interacting system via its particle density and not via its many-body wave function. Its significance is to reduce the 3N degrees of freedom of the N-body system to only three spatial coordinates through its particle density. The original DFT has been generalized to deal with many different situations; spin polarized systems, multicomponent system such as nuclei and electron hole droplets free energy at finite temperature, superconductors with electroni[4].c pairing

mechanisms, relativistic electrons, time dependent phenomenon and excited states, bosons, molecular dynamics etc.

Electron density is the central quantity in DFT. The probability of finding any of the N electrons within the volume element $d\mathbf{r}$ is given by,

$$\rho(\mathbf{r}) = N \int .... \int |\psi(\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_N)|^2 d\mathbf{x}_1 d\mathbf{x}_2 ... d\mathbf{x}_N \tag{2.7}$$

where, $\rho(\mathbf{r})$ is a non-negative function of only the 3 spatial variables which vanishes at infinity and integrates to the total no. of electrons:

$$\rho(\mathbf{r}) \leqslant 0$$

$$\rho(\mathbf{r} \to \infty) = 0$$

$$\int \rho(\mathbf{r}) d\mathbf{r} = N$$

$\rho(\mathbf{r})$ is an observable and can be measured experimentally, e.g. by X-ray diffraction.

2.0.4 Thomas-Fermi-Dirac Approximation

The old DFT was Thomas-Fermi (TF) model proposed by Thomas and Fermi in 1927. In this method, they used the electron density $n(\mathbf{r})$ as the basic variable instead of the wave function. The external potential $V_{ext}(\mathbf{r})$ is written as a function of the electron density $n(\mathbf{r})$ as,

$$E_{TF}[n(\mathbf{r})] = A_1 \int n(\mathbf{r})^{5/3} d\mathbf{r} + \int n(\mathbf{r}) V_{ext}(\mathbf{r}) d\mathbf{r} + \frac{1}{2} \int \int \frac{n(\mathbf{r}) n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \tag{2.8}$$

where, the first term is K.E. of the non-interacting electrons in a homogeneous electron gas (HEG) with $A_1 = \frac{3}{10}(3\pi^2)^{2/3}$ in atomic units. The second term is the classical electrostatic energy of the nucleus-electron coulomb interaction and third term is the classical electrostatic Hartree energy approximated by the classical coulomb repulsion between electrons. In the original TF method, the exchange and correlation among electrons was neglected. In 1930, Dirac extended the TF method by adding a local exchange term, $A_2 \int n(\mathbf{r})^{4/3} d\mathbf{r}$ to equation (2.8) with $A_2 = -\frac{3}{4}(3/\pi)^{1/3}$.

Then equation (2.8) becomes,

$$E_{TFD}[n(\mathbf{r})] = A_1 \int n(\mathbf{r})^{5/3} d\mathbf{r} + \int n(\mathbf{r}) V_{ext}(\mathbf{r}) d\mathbf{r} + \frac{1}{2} \int \int \frac{n(\mathbf{r}) n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' + A_2 \int n(\mathbf{r})^{4/3} dr \tag{2.9}$$

The ground state density and energy can be obtained by minimizing TFD equation. By using the technique of Lagrangian multipliers, then TFD equation becomes,

$$\frac{5}{3}n(\mathbf{r})^{2/3} + V_{ext}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' + \frac{4}{3}A_2 n(\mathbf{r})^{1/3} - \mu = 0 \tag{2.10}$$

which have solved the directly to obtain the ground state density.
This theory fails to describe bonding between atoms, thus molecules and solids cannot form in this theory [36]. Although it is not good enough to describe electrons in matter, its concept to use electron density as the basic variable illustrates the way DFT works.

2.0.5 Hohenberg-Kohn Theorem

This theorems published by the P. Hohenberg and W. Kohn in 1964 called as Hohenberg-Kohn (HK) theorems. HK theorem states the ground state wave function of a many particle system, if non-degenerate is unique functional of the particle density [37].
For any system of interacting in an external potential $V(\mathbf{r})$, the external potential is uniquely determined (except for a constant) by the ground state density $n_0(\mathbf{r})$.

First Hohenberg-Kohn (HK) theorem

This theorem states that, the external potential $V_{ext}(\mathbf{r})$ is (to with in constant) a unique function of $\rho(\mathbf{r})$. Then we have to show that two different ground state wave functions cannot lead to the same ground state density.
Let two different systems with ground state wave function ψ_0 and ψ_0'. The K.E. operator for the N-electron system is *'T'*, the operator that describe interaction is *'W'* and $V = \Sigma v(\mathbf{r}_i)$ are the external potential of the two systems. If we consider,

$$V(\mathbf{r}) \neq V'(\mathbf{r}) + c$$

where, c is constant.
Then,

$$H\psi_0 = (T + W + V)\psi_0 = E_0\psi_0 \tag{2.11}$$

$$H'\psi_0' = (T + W + V)\psi_0' = E_0'\psi_0' \tag{2.12}$$

Starting with the two system described by equation (11) and (12),

$$H = H' + V - V'$$

The ground of system 1 is,

$$E_0 = \langle \psi_0 | H | \psi_0 \rangle$$

From the variational theorem this must be less than expectation value, $\langle \psi_0 | H | \psi_0 \rangle$ because the wave function, ψ_0' is not the lowest eigenstate of H, i.e. ψ_0' can not the correctly describe the ground state of system 1.
Therefore,

$$E_0 < \langle \psi_0' | H | \psi_0' \rangle$$

$$E_0 < \langle \psi_0' | H' + V - V' | \psi_0' \rangle$$

$$E_0 < \langle \psi_0' | H' | \psi_0' \rangle + \langle \psi_0' | v - v' | \psi_0' \rangle$$

$$E_0 < E_0' + \int n_0'(\mathbf{r})[V(\mathbf{r}) - V'(\mathbf{r})]d\mathbf{r} \tag{2.13}$$

where, $\langle \psi_0' | H | \psi_0' \rangle = \int n_0'(\mathbf{r}) V(\mathbf{r}) d\mathbf{r}$ $n_0(\mathbf{r}) = \int |\psi(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N)|^2 d\mathbf{r}_1 d\mathbf{r}_2 .... d\mathbf{r}_N$
Similarly, from the variational theorem, we also know that the ground state energy E_0' of the system 2 must be less than the expectation value of $\langle \psi_0' | H' | \psi_0' \rangle$,

$$\therefore E_0' < E_0 + \int n_0(\mathbf{r})[V'(\mathbf{r}) - V(\mathbf{r})]d\mathbf{r} \tag{2.14}$$

If we assume, ψ_0 and ψ_0' can lead to the same ground state density we have that $n_0(\mathbf{r}) = n_0'(\mathbf{r})$.
Then,

$$E_0 + E_0' < E_0' + E_0$$

Therefore, two different ground state wavefunction can not lead to the same ground state density.

Second Hohenberg-Kohn (HK) theorem

This theorem states that, two different potentials can not produce the same ground state wave function.

If the two systems described by different potentials, V and V′, lead to the same ground state wave function $\psi_0 = \psi_0'$.

Then,

$$(V - V')\psi_0 = (E_0 - E_0')\psi_0$$

Then, clear contraction to the requirement that the potential be different,

$$(V - V') = (E_0 - E_0')$$

Thus, if the ground state wave function of two system are the same, the potential that describe those systems can only differ by a constant.

Thus, two different potential can not produce the same ground state wave function.

2.0.6 Kohn-Sham Theorem

The Kohn-Sham equation is the Schrodinger equation of a fictitious system of non-interacting particles (typically electrons) that generate the same density as any given system of interacting particles [38]. It is much easier to solve than the coupled Schrodinger equation. The great advantage of the KS equation over Thomas-Fermi theory is that almost all the K.E. (T_S) is treated exactly.

We have, the ground state energy of the system can be written as,

$$E_0 = min_{\rho \to N} \left[E(\rho) + \int \rho(\mathbf{r}) V_{Ne} d\mathbf{r} \right] \tag{2.15}$$

where, $E(\rho)$ = Universal function contains of the K.E., the classical coloumb interaction and non-classical portion,

$$E(\rho) = T(\rho) + J(\rho) + E_{ncl}(\rho) \tag{2.16}$$

only $J(\rho)$ is know, the main problem is to find the $T(\rho)$ and $E_{ncl}(\rho)$. To solve this problem KS proposed in 1965 the approach described below. To calculate the exact K.E. of a non-interacting reference system with the same density as the real, interacting one,

$$T_s = -\frac{1}{2} \sum_i^N \langle \psi_i | \bigtriangledown^2 | \psi_i \rangle$$

$$\rho_s(\mathbf{r}) = \sum_i^N \sum_s |\psi_i(\mathbf{r}, s)|^2 = \rho(\mathbf{r}) \tag{2.17}$$

where, ψ_i = orbital of the non-interacting system.

T_s is not equal to the true K.E. of the system. Kohn and Sham accounted for that by introducing the following separation of the functional $E(\rho)$,

$$E[\rho] = T_s(\rho) + J(\rho) + E_{XC}(\rho) \tag{2.18}$$

where, E_{XC} = exchange-correlation energy

$$E_{XC} = (T(\rho) - T_s(\rho) + (E_{ee}(\rho) - J(\rho)) \tag{2.19}$$

Equation (18) can be written as,

$$E[\rho] = T_s(\rho) + J(\rho) + E_{XC}(\rho) + E_{Ne} \tag{2.20}$$

$$E[\rho] = T_s(\rho) + \frac{1}{2}\int\int \frac{\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)}{\mathbf{r}_{12}} d\mathbf{r}_1 d\mathbf{r}_2 + E_{XC}(\rho) + \int V_{Ne}\rho(\mathbf{r})d(\mathbf{r})$$

$$E(\rho) = -\frac{1}{2}\sum_{i=1}^N \langle\psi_i| \bigtriangledown^2 |\psi_i\rangle + \frac{1}{2}\sum_i^N\sum_i^N \int\int |\psi_i(\mathbf{r}_1)|^2 \frac{1}{\mathbf{r}_{12}} |\psi_j(\mathbf{r}_2)|^2 d(\mathbf{r}_1)d(\mathbf{r}_2) + E_{XC}(\rho)$$
$$-\sum_i^N \int \sum_A^M \frac{Z_A}{\mathbf{R}_{1A}} |\psi_i(\mathbf{r}_1)|^2 d\mathbf{r}_1 \tag{2.21}$$

The only term for which no explicit form can be given as E_{XC}. The resulting equation are KS equation,

$$\left(-\frac{1}{2}\bigtriangledown^2 + \int \frac{\rho(\mathbf{r}_2)}{\mathbf{r}_{12}} d\mathbf{r}_2 + V_{xc}(\mathbf{r}_1) - \sum_A^M \frac{Z_A}{\mathbf{r}_{1A}}\right)\psi_i = \left(-\frac{1}{2}\bigtriangledown^2 + V_s(\mathbf{r}_1)\right)\psi_i = \epsilon_i\psi_i \tag{2.22}$$

$$V_s(\mathbf{r}_1) = \int \frac{\rho(\mathbf{r}_2)}{\mathbf{r}_{12}} d\mathbf{r}_2 + V_{xc}(\mathbf{r}_1) - \sum_A^M \frac{Z_A}{\mathbf{r}_{1A}} \tag{2.23}$$

where, $V_{XC} = \dfrac{\delta E_{XC}}{\delta\rho}$

It is very important to realize that if exact form of E_{XC} and V_{XC} are known, the KS strategy would lead to the exact energy.

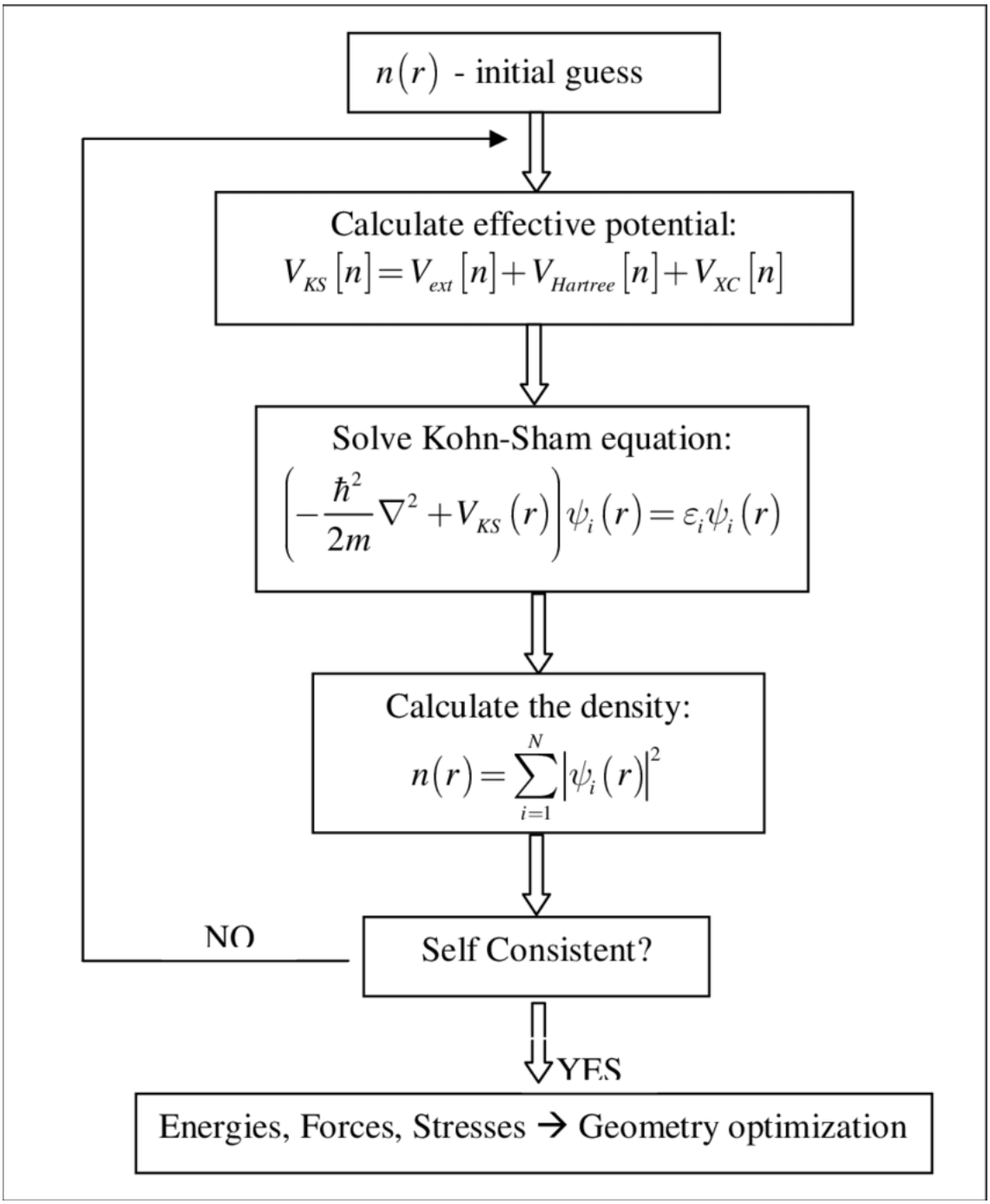

Figure 2.1: Self-consistency flowchart showing working procedure in KS formalism [39].

2.0.7 Exchange-Correlation Functional

The main problem with the Kohn-Sham procedure described above is the introduction of an unknown exchange-correlation functional. As long as there is no expression for this term, it is impossible to solve the KS equations. The central goal of modern density functional theory is therefore to find better and better approximations to these two quantities. Therefore, several approximations have been devised to get an explicit form for this functional. The most widely used approximation in solid state physics are the local density approximation (LDA) and the generalized gradient approximation (GGA).

Local Density Approximation(LDA)

The Kohn-Sham scheme constitutes a practical way to implement density functional theory, provided an accurate and reasonably easy to use approximation is available for the exchange-correlation energy E_{xc}. Local density theories based on the HK theorem have been applied to atoms and molecules. The major problem of DFT is that the exact functionals for exchange and correlations are not known except for free electron gas. But, some approximation exists like local density approximation (LDA), which is most widely used approximation. The approximation permit the calculation of certain physical quantities quite accurately. LDA assumes, density is same everywhere.
The LDA is the basis of all approximate exchange-correlation functionals. At the center of this model is the idea of an uniform electron gas. The central idea of LDA is the assumption that we can write E_{xc} in the following form,

$$E_{xc}^{LDA}[\rho] = \int \rho(\mathbf{r})\epsilon_{xc}(\rho(\mathbf{r}))d\mathbf{r} \tag{2.24}$$

Here, $\epsilon_{xc}(\rho(\mathbf{r}))$ is exchange-correlation energy per particle of an uniform electron gas of density $\rho(\mathbf{r})$. Taking the functional derivative of E_{xc}^{LDA}, we find exchange-correlation potential as

$$v_{xc}^{LDA} = \frac{\delta E_{xc}^{LDA}}{\delta\rho} = \epsilon_{xc}[\rho(\mathbf{r})] + \rho(\mathbf{r})\frac{\partial\epsilon_{xc}[\rho]}{\partial\rho} \tag{2.25}$$

The exchange-correlation energy per particle $\epsilon_{xc}(\rho(\mathbf{r}))$ can be further split into exchange and correlation contributions,

$$\epsilon_{xc}(\rho(\mathbf{r})) = \epsilon_x(\rho(\mathbf{r})) + \epsilon_c(\rho(\mathbf{r})) \tag{2.26}$$

Now, equation (2.24) can be written as

$$E_{xc}^{LDA}[\rho] = \int \rho(\mathbf{r})[\epsilon_x(\rho(\mathbf{r})) + \epsilon_c(\rho(\mathbf{r}))] = E_x[\rho(\mathbf{r})] + E_c[\rho(\mathbf{r})] \tag{2.27}$$

For spin polarized systems,

$$E_{xc}^{LSDA}[\rho_\uparrow(\mathbf{r}), \rho_\downarrow(\mathbf{r})] = \int \rho(\mathbf{r})\epsilon_{xc}[\rho_\uparrow(\mathbf{r}), \rho_\downarrow(\mathbf{r})]d\mathbf{r} = \int \rho(\mathbf{r})[\epsilon_x[\rho_\uparrow(\mathbf{r}), \rho_\downarrow(\mathbf{r})]+\epsilon_c[\rho_\uparrow(\mathbf{r}), \rho_\downarrow(\mathbf{r})]]d\mathbf{r} \tag{2.28}$$

The exchange term $\epsilon_x[\rho]$ is first derived originally by Bloch and Dirac which is given by atomic units as,

$$\epsilon_x[\rho] = -\frac{0.458}{r_s} \tag{2.29}$$

Where, r_s is the radius of a sphere containing one electron and is given by, $(4\pi/3)r_s^3 = \rho^{-1}$. The correlated part is first estimated by E. P. Wigner and is given as

$$\epsilon_c[\rho] = -\frac{0.44}{r_s + 7.8} \tag{2.30}$$

The accuracy of LDA is often considered satisfactory in condensed matter system, but it is much less so in atomic and molecular physics, for which highly accurate experimental data are available. For the case of cohesive energy and bond strengths in molecules and solids LDA overestimates, so this problem is corrected by introducing gradient corrections, i.e. GGA.

Generalized Gradient Approximation(GGA)

Generalized gradient approximation (GGA) is also most widely used approximation in solid state physics [40]. GGA can improve upon the local spin density approximation (LSDA) description of atoms, molecules and solids. GGA's give reliable results for all main types of chemical bonds (covalent, ionic, metallic and hydrogen bridge). But for Vander-Waals interactions both GGA and LDA are fail to explain. So, GGA's has lead to major improvements as compared to LDA in both physics and chemistry.

To correct for LDA's density tendency, it is common to expand in terms of gradient of the density for the non-homogeniety of the true electron density. In GGA, gradient correlation are added. The exchange-correlation energy of GGA can be expressed as,

$$E_{XC}^{GGA} = \int \epsilon_{xc}(\rho(\mathbf{r})| \bigtriangledown \rho(\mathbf{r})|, ...)d\mathbf{r} \tag{2.31}$$

For some systems GGA improves on LDA but for some system GGA fails, LDA lead to correct results.

Potentially meta-GGA is more accurate than the GGA Functionals. Meta-GGA DFT functional in its form includes the second derivatives of the electron density (the Laplacian), but GGA includes only the first derivative and density in the exchange-correlation potential.

2.1 Green's Function (GF)

The first section of this chapter briefly reviewed how DFT can be used to obtain the Hamiltonian of the system. In this section I will demonstrate the way to use the DFT mean field Hamiltonian (DFT-mfH) via Greens function method to calculate the transmission coefficient $t(E)$ [27, 28]. Therefore the GFs procedures followed to calculate $t(E)$ are;

- Calculating the GF of double infinite periodic lead.
- Calculating surface GF of the lead via applying boundary conditions.
- Using Greens function of the whole system, i.e., a system with two leads plus scatter region.
- Calculating the $t(E)$ of the electron from one lead to the other through the molecule.

2.2 Tight binding (TB)

Besides DFT, a number of methods are available which can fulfill the same task, with different accuracy. One of these methods is the Tight Binding method. The elegancy of the TB is the localization of the potentials and the wavefunctions, presumed to be orthonormal, are centered on the atomic sites [30]. These properties essentially result in a parameter dependent Hamiltonian,. Furthermore, these parameters can be extracted through fitting to empirical, semi-empirical, DFT, or ansatz calculations [30]. In addition to the simplicity, TB is a very effective tool to explain and analyse the behaviour of materials and obtain the transport properties.

2.3 Pseudopotential

One of the main advantages of using a plane wave basis set is that its accuracy can be easily controlled. This is related to the fact that, when using such a basis set, we are making no assumptions about the final shape of the orbitals, other than that there is some scale below which they become smoothly varying. However, this also leads

to a major disadvantage of using a plane wave basis set, which is that the size of the basis set required for a given system is often far larger than would be required with a localised basis set. This is because, in condensed matter systems, the orbitals tend to oscillate very rapidly in the vicinity of atomic nuclei, and are much more smoothly varying elsewhere. In order to describe this rapid oscillation we must set a very large cut-off energy, so that we include plane waves with very short wavelengths. But, since most of the space in the cell does not contain rapidly oscillating orbitals, most of the computational expense associated with all these plane waves effectively goes to waste. A localised basis set can be tailored such that the basis functions themselves are rapidly oscillating in the vicinity of atomic nuclei and more smoothly oscillating elsewhere, so that the total number of basis functions required for the system is far smaller. For this particular Purpose we have used ATOM program to generate required pseudopotetial.[36]

Chapter 3

Details of Simulation

3.1 General Consideration

3.2 Computational codes and DFT

Today many open source and proprietary computer codes are available which provide a way to compute and study various properties of materials using density functional theory. These codes differ in terms of their implementations used to solve KS equations.In our study of two dimensional gold monolayer we have used SIESTA *(Spanish Initiative for Electronic Simulations on Thousands of Atoms)* code [42] which is an open source code, based on numerical atomic orbitals as basis set and is very efficient, easy to run on a quad core machine in parallel mode. In the following, a brief description of SIESTA code, which we are using to calculate the structural, electronic and transport properties of transition metal gold (monolayer) using DFT, has been discussed.

3.2.1 SIESTA Code

SIESTA (Spanish Initiative for Electronic Simulations on Thousands of Atoms) is a calculation method and computer code which solves problems of condensed matter using density functional theory (DFT) where most of the problems are related to ground state properties like energy/volume curves, band structure and phonons etc. SIESTA uses the standard KS self-consistent density functional method in the local density (LDA) or generalized gradient (GGA) approximations with norm-conserving pseudopotentials [50]. SIESTA solves the Kohn-Sham equations by expanding the wave functions

using localized, numerical atomic orbitals as basis set, which is the main characteristic of SIESTA. It can be implemented for larger systems of thousands of atoms . SIESTA is very good for study of low dimensional materials, because presence of vacuum does not involve an extra cost as in the case of plane-wave approaches. The efficiency and the wide-range of utilities indeed make SIESTA a very useful tool for the study of complex materials. It can be used to study the structural properties, electronic properties, magnetic properties, optical properties and transport properties of materials containing thousand of atoms. The first task to study any property of material is the structure optimization. In SIESTA structural optimization is done by conjugate gradients(CG). The relaxation method moves the atoms trying to minimize the forces on them. After structural relaxation, we can determine the other properties accurately.

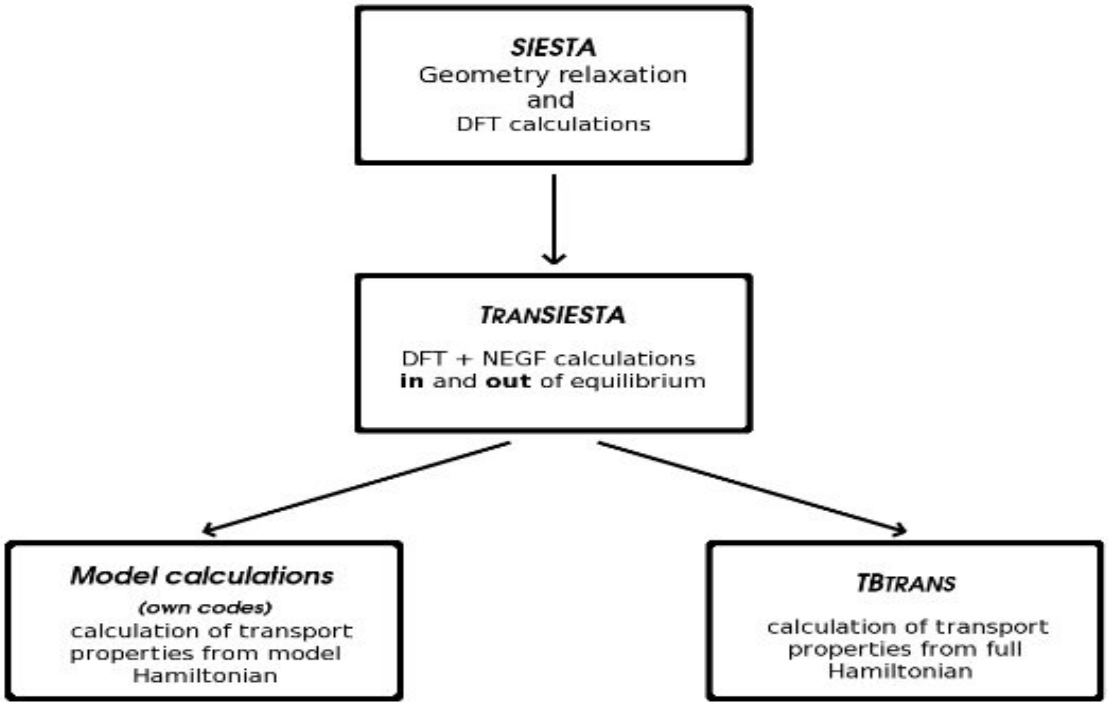

Figure 3.1: Structure chart of the transport calculations within our work.

We use for the first-principles calculations the DFT code SIESTA, its DFT+NEGF code TRANSIESTA and the post-processing tool TBTRANS. Using SIESTA, we relax the structures of the systems and calculate the electron density self-consistently, once for the central region and once for the electrodes. Using TRANSIESTA, we calculate the electron density of the transport system from the results of the SIESTA calcula-

tions. Using TBTRANS , we calculate the transport properties of the systems from the results of the TRANSIESTA calculations. Additionally, we reduce the Hamiltonian matrices gained from the TRANSIESTA calculations to model Hamiltonian matrices, from which we calculate transport properties allowing further insight.

3.2.2 TRANSIESTA

Using SIESTA only systems of finite size or periodic systems can be calculated. Therefore, SIESTA does not allow the treatment of typical transport systems being infinite and non- periodic. We calculate transport systems in and out of equilibrium using TRANSIESTA [47]

Electrodes and Central region

The transport system is split into three parts: the two electrodes and the central region. The semi-infinite electrodes are further split into identical clusters and treated using the concept of surface Greens functions [49] where the surface Greens functions are calculated using the Sancho-Rubio method where the electrode clusters are allowed to couple only to nearest-neighboring ones. Therefore the cluster size in transport direction and the cutoff radius of the electrode wavefunctions cannot be chosen independently from one another. The partitioning of the system into electrodes and central region in TRANSIESTA differs slightly from the partition. The Hamiltonian matrices and overlap matrices of the electrode clusters and of the central region required to deal with the transport system using the DFT+NEGF approach are gained from SIESTA calculations.

3.2.3 TBTRANS

TBTRANS [47] is a post-processing code allowing the analysis of TB Hamiltonian matrices. We will use it to analyze the results gained from the TRANSIESTA calculations. We will calcu- late a variety of properties using TBTRANS : the transmission, which is calculated the current, and the DOS of the central region.TBTRANS allows the calculation of the DOS of the central region either by the spectral function or by the spectral functions of the left and right electrode from, where the latter approach is more suited for transport systems, especially in non-equilibrium.

3.2.4 Parameters considered in the SIESTA and TRANSIESTA calculations

SIESTA and TRANSIESTA allow a variety of options and parameters to be set. Due to the vast amount of these parameters we discuss only the most important ones.

K-points and MeshCutoff

The k-points determine the fineness of the grid on the unit cell or supercell specified by the user.Different numbers of k-points also result in different k-points used for calculations. Thus for a non-converged grid no reliable results can be obtained neither can these results be used for interpolation. The parameter MeshCutoff defines the fineness of the grid in real space. The same considerations as for the grid in k-space are valid.

BasisSize, EnergyShift and SplitNorm

A too small basis will restrict calculations, e.g. geometry relaxation, to much and will result in a bad description of the system. Therefore the basis size has to be converged. The basis size can be specified by SZ, DZ, SZP or DZP. SZ and DZ specify the usage of one basis function (single zeta) or two basis functions (double zeta) per orbital, respectively, where the basis functions are not limited to being linear combinations of Gaussians. The basis sizes SZP (SZ + polarization) and DZP (DZ + polarization) consider additional polarization orbitals. The parameter EnergyShift determines the cutoff radii. Forces are only present up to the cutoff radius. A too short cutoff radius can therefore yield to wrong results due to the absence of long-range forces. Also the transmission can be wrong due to too many entries in the overlap matrix and in the Hamiltonian matrix being zero. The parameter SplitNorm determines how much of the norm of the Slater type orbital goes into the second-zeta function. A too low or too high value will result in the atom orbitals being described mainly by one of the two basis functions.

MD.MaxForceTol

MD.MaxForceTol defines the maximum force below which the geometry of the system is con- sidered to be relaxed during structure optimization. A too large force tolerance

will cause the geometry optimization to finish before the structure is sufficiently relaxed. A too small force tolerance increases the computation time.

Contour

TRANSIESTA calculates the equilibrium density matrix from the Greens function using a complex contour integral. The Greens function is smoother in the complex plane, which allows an accurate evaluation with less sampling points and also the implementation of numerically efficient methods. In the complex plane one can also take advantage of the residue theorem

Chapter 4

Results and Discussion

4.1 General Consideration

In this chapter we outline and explain the computational process of first-principles study carried out to investigate the structural, and electronic properties of Gold monolayer. The calculations are performed in the framework of the Density Functional Theory(DFT) with van der Waals interaction under the generalized gradient approximation (GGA) of Perdew, Burke, and Ernzerhof (PBE) (to account electronic exchange and correlation effects) and self consistent plane wave pseudo potential,implemented with the SIESTA and TRANSIESTA Pacakage.

The pseudo potential approach is an attempt to replace the complicated effects of the motion of the core (i.e.non-valence) electrons of an atom and its nucleus with an effective potential or pseudo potential and only the chemically active valence electrons are included explicitly in calculation. The inner cores is replaced by pseudo potential. As it is already tested that core electron has no role in chemical bonding and reaction.

While in GGA method using SIESTA Code, the kinetic cut-off energy, suitable values of k-points and lattice optimization is done before variable cell relaxation and scf calculations. After the calculation of three basic parameters; meshcutoff energy, k-point, and lattice parameter we constructed a unit cell using these parameters in input file. After the convergence test we obtained the basic parameters for the input file. The optimized geometry of gold bulk is obtained from relax calculation and total energy is obtained from scf calculation. From the information of relax and scf output files we

tested the stability of system and obtained their geometrical structures.

4.2 Computational Detail

Studies start with the structural relaxation with and without spin polarization. In the process the electronic states have been expanded in a linear combination of localized numerical pseudoatomic orbitals (LCAOs) using a split valence double-zeta polarized (DZP)-basis set for the valence electrons. TrouillerMartins norm-conserving pseudopotentials were used to describe the interactions between core and valence electrons. The exchange-correlation energies have been treated with the Perdew-Burke-Ernzerhof (PBE) functional in the generalized gradient approximation (GGA): 25×1 × 25 MonkhorstPack grid, for each unit cell, and an energy cutoff of 375 Ry for the grid integration have been applied. The electron temperature was set equal to 300K with a Fermi surface broadening. The atomic relaxation was achieved when the force reached the value 20 meV, following the Hellmann-Feynman theorem, and the electronic relaxation was converged to 1.0×10^{-4}. A vacuum gap of 13 A between adjacent chains in the lateral (x and z) directions has been used to prevent interactions between adjacent unit cell.

4.3 Convergence Test

Before calculating the electron density of states, band structure etc, it is advisable to check the convergence of the system by calculating the total energy versus mesh cutoff and total energy versus k-points. Different mesh cut-off energies raging from 100 Ry to 500 Ry have been tested for nano-structure of gold monolayer containing 45 atoms. The mesh cutoff 350 Ry was found to be suitable for all calculations.
K-points are set according to the Monkhorst Pack scheme . Different values starting from 10 to 30 k-points have been tested for the monolayer and an $25 \times 1 \times 25$ mesh has been found suitable for all calculations with minimum value of energy, because it gives good convergence at a reasonable computational cost.

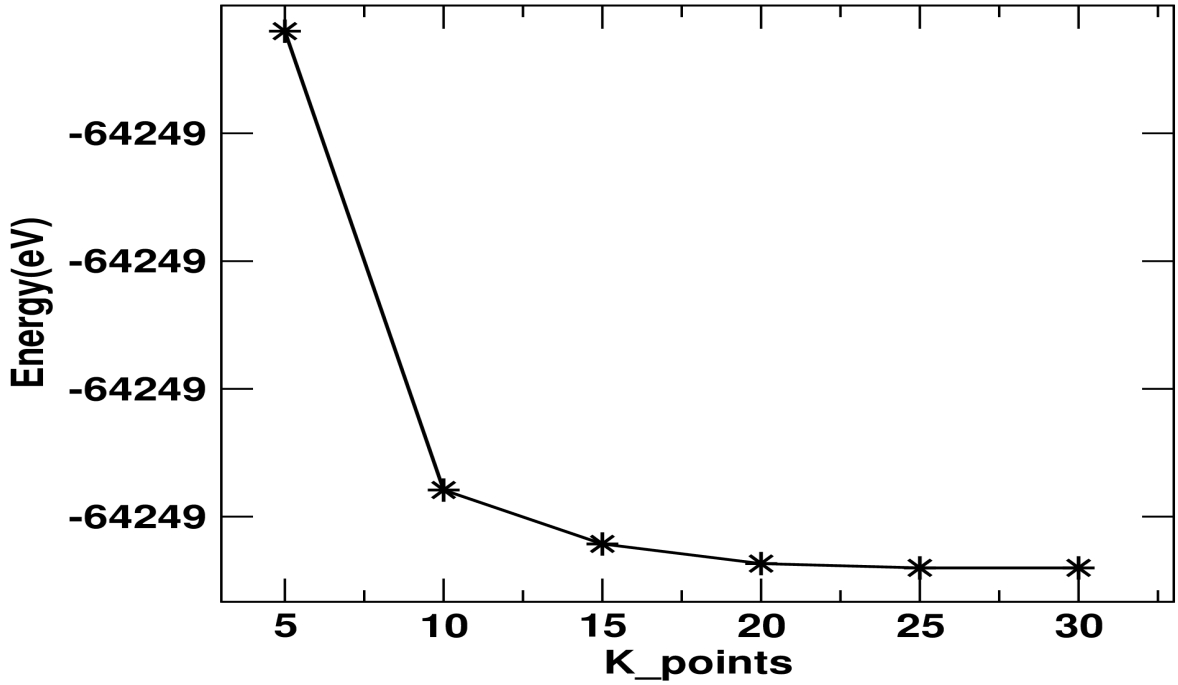

Figure 4.1: Convergence of computed total energy vs k point for gold monolayer

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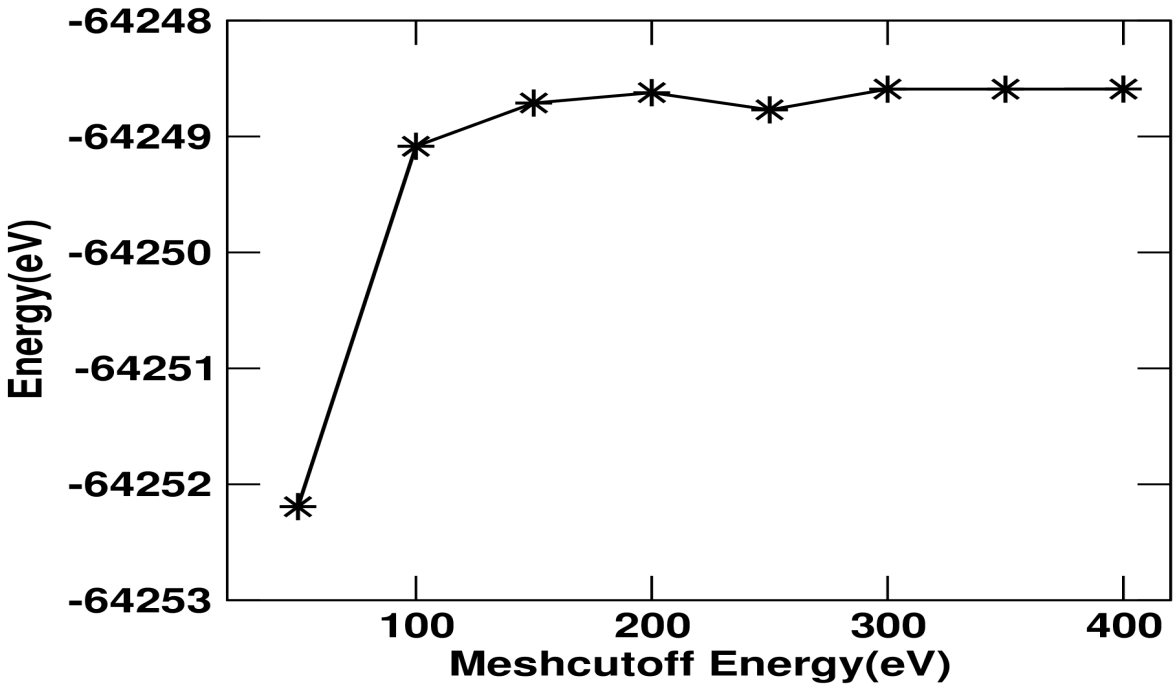

Figure 4.2: Convergence of computed total energy vs mesh cutoff energy for gold monolayer .

4.4 Structural Properties

At first, the unit cell (u.c) for the structure, with and without spin effect, has been optimized. Null total magnetic moments in the spin-polarized calculations have been found, and therefore Au monolayer configuration, non-magnetic were confirmed. Fig. 1(a) shows the two dimensional monolayer is along the xz plane and a large vacum of 13 A was provided on the adjacent side of the y direction to remove the interaction.The transport direction is always fixed parallel to the a the z direction in the code.In the system configuration the bond distances are found to be 2.65 A and 2.7 A respectively after the geomertric optimization and relaxation.

Table 4.1: Calculated different parameter of gold monolayer and bulk structure

S.N	Properties)	Gold monolayer	Gold bulk	Exp. Bulk
1	Bond length	a=2.65,b=2.70	2.60A	2.70 A
2	Cohesive energy	2.98 ev/atom	3.79 eV/atom	3.81ev/atom
3	No.of Bonds/atom	4	12	
4	Bulk Modulus		170 GPA	173.2GPA

4.4.1 Cohesive energy of the 2D Au monolayer

To evaluate the stability of this structure, we first computed the cohesive energies $E_{\text{coh, monolayer}} = (mE_{\text{Au atom}} - E_{\text{Au monolayer}})/m$, and $E_{\text{coh bulk}} = (mE_{\text{Au atom}} - E_{\text{Aubulk}})/m$. With $E_{\text{Auatom,}}$ $E_{\text{Au-monolayer,}}$ and E_{Aubulk} being the total energies of a single Au atom, one unit cell of the Au monolayer and one unit cell of bulk gold in face-centered cubic (fcc).The Au monolayer and bulk has a cohesive energy of 2.98 eV/atom and 3.79 eV per atom respectively.Due to the less number of bonds per atom in the 2D layer compared to the 3D bulk we observe significantly enhanced energy per bond . This is similar to the increase in bond strength going from 3D diamond to 2D graphene. It is a non-magnetic metal, and was found to be the global minima in the 2D space.The high cohesive energy reveal strong bonds in the monolayer. This suggest that this monolayer is very stable after the formation although the layer has not been fabricated yet but they can be grown on a surface which may be stabilized by the substrate.

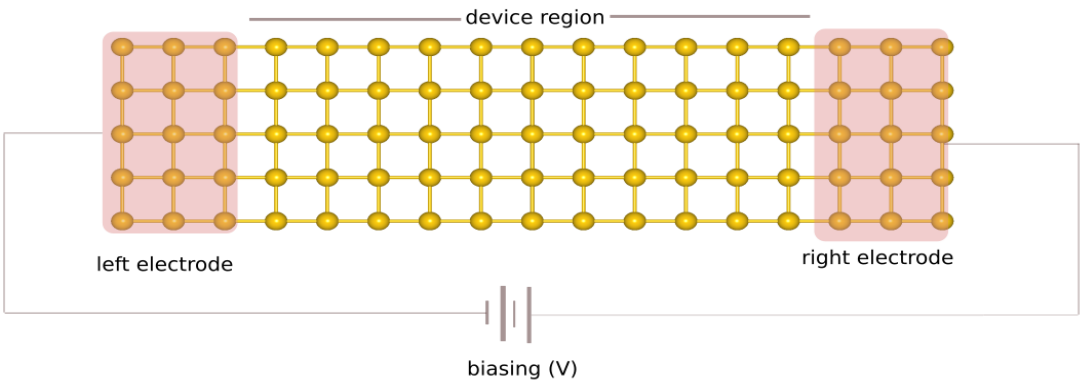

Figure 4.3: Gold monolayer and the transport biasing module .

Properties	gold (Au)
Atomic No.	79
Electronic Configuration	[Xe] $5f^1 4\ 5d^1 0\ 6s^1$
XC. Functional	GGA
XC. Authors	PBE
Atomic calculation	Relativistic
R_c **(a.u.)**	6s = 2.29,6p = 3.14,5d=1.20,5f=3.14
Core correction	Non-linear
Type of Pseudization	Troullier-Martins

4.5 Electronic Properties

To get insight into electronic properties, we have computed the band structure as well as its density of states (DOS). As shown in Fig. 4.4, the material shows a band structure typical of metals.The band structure and the orbital associated with bonding and conduction properties are described in an energy range of the vicinity of fermi level is depicted. The metallic character of the Au monolayer is demonstrated by the Fermi level located inside the bands, and no band gap was observed at this energy. We see that there is large number of bands at Valence band region in the energy range from -1 eV to -2 eV and less bands are on the Conduction band , similar to the bulk.

The partial density of states (PDOS) analysis shows that below the Fermi energy

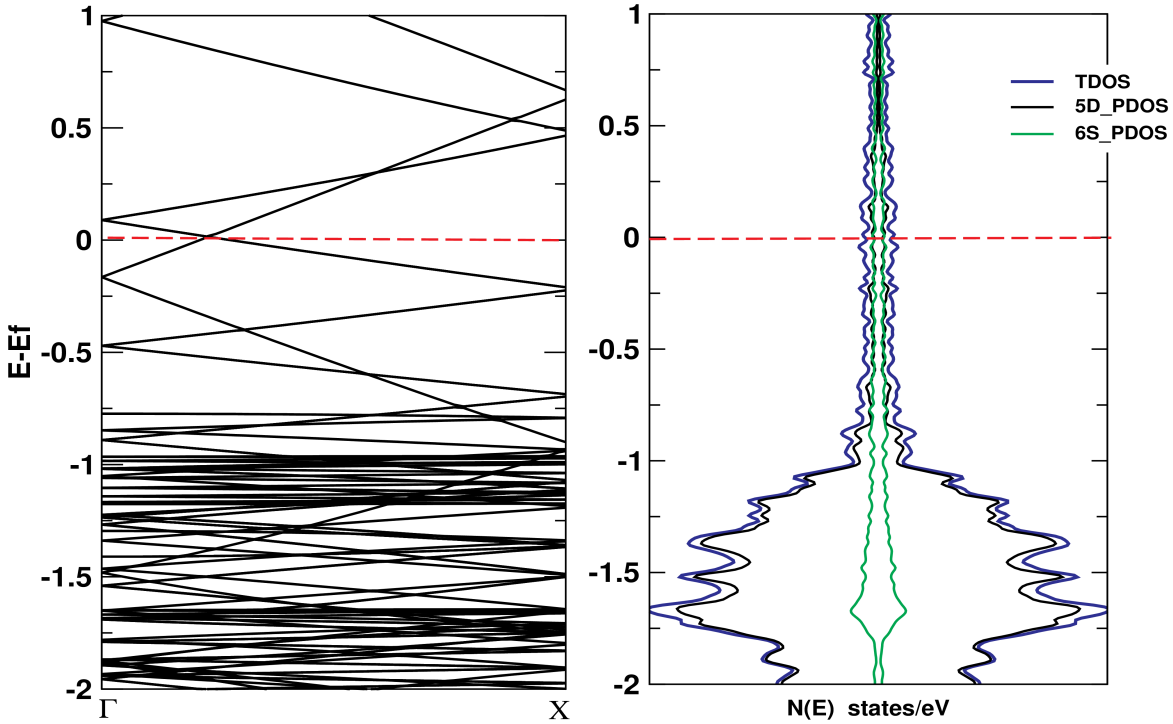

Figure 4.4: The band structure of gold monolayer and (b) Spin dependent total density of state (TDOS) and partial density of states (PDOS) of constitnent atoms .

the major contribution comes from Au 5d states where as other orbital are not so involving. The symmerty in spin polarized calculation shows that gold monlolayer is purely metallic in nature as the bulk structure. The band structure indicates that these materials are metallic with a large density-of-states (DOS) at the Fermi level. The Au monolayer is diamagnetic as confirmed by a spin-polarized calculation. similary, fig 4.5 depicts the electronic band structure and density of states of gold bulk where the 5d orbitals is the major contributor.

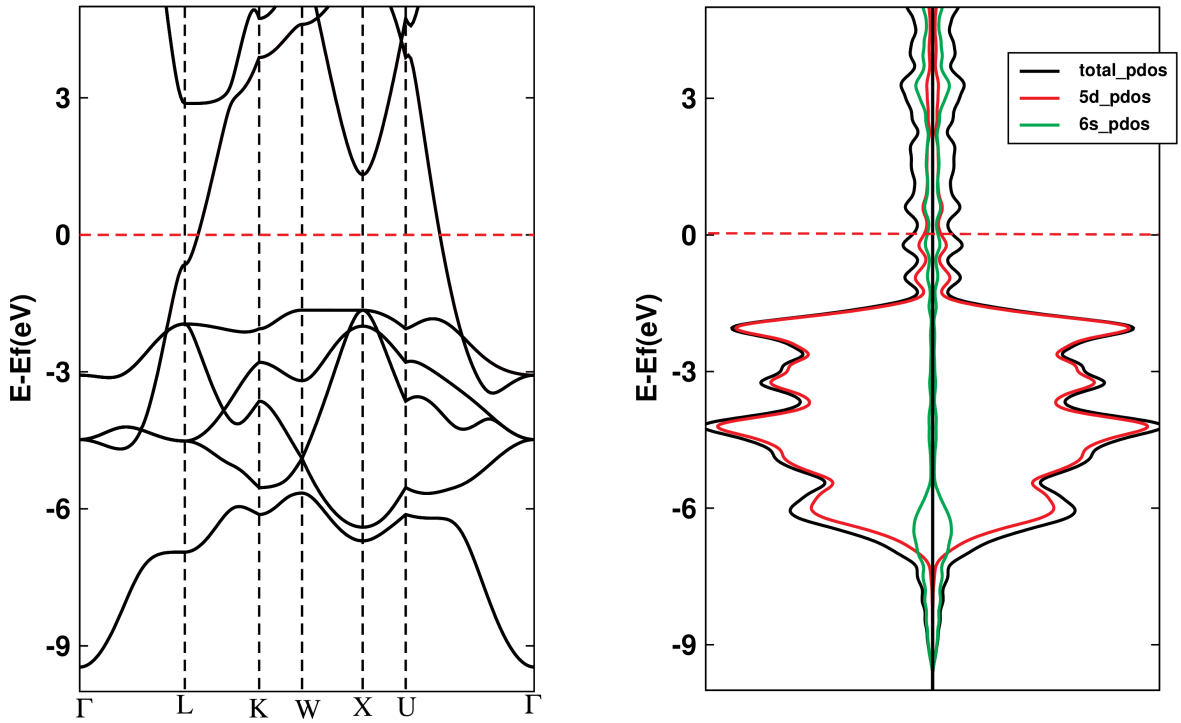

Figure 4.5: The band structure of gold bulk and (b) Spin dependent total density of state (TDOS) and partial density of states (PDOS) of constitnent atoms .

4.6 Transport Properties

The electronic transport properties have been studied with Non Equilibrium Greens Function (NEGF) techniques as implemented in Transiesta [13-16] package based on density functional theory. In the gold monolayer 8 atoms of each sides are taken as the electrode and 27 atoms are considered as the device region at which scattering takes place resulting ballastic transport from left to right electrode. The electronic transport properties of the optimized systems have been calculated with the nonequilibrium Green's function technique (NEGF) as implemented in the TransIESTA module. We determine the current through the scattering region sandwiched between two semi-infinite electrodes at a finite bias voltage (V_{b}) using the Landauer-Bttiker formula,

$$\mathrm{I}(V_{\mathrm{b}}) = G_0 \int_{\mu_{\mathrm{L}}}^{\mu_{\mathrm{R}}} T\left(E, V_{\mathrm{b}}\right)\left[f\left(E, \mu_{\mathrm{L}}\right) - f\left(E, \mu_{\mathrm{R}}\right)\right] \mathrm{d}E \tag{4.1}$$

here f is the Fermi distribution function, $\mu_{\mathrm{L,R}}$ are the chemical potentials of the left and right electrodes with $\mu_{\mathrm{L}} = E_{\mathrm{F}} + V_{\mathrm{b}}/2$ and $\mu_{\mathrm{R}} = E_{\mathrm{F}} - V_{\mathrm{b}}/2$ shifted up and down relative to the E_{F} level; and $T\left(E, V_b\right)$ is the transmission coefficient at energy E and

bias voltage $V_{\mathrm{b}} = (\mu_{\mathrm{L}} - \mu_{\mathrm{R}})/e$. The electron transmission coefficient from the left to the right electrodes can be calculated by

$$T(E, V_{\mathrm{b}}) = \mathrm{Tr}\left[\Gamma_{\mathrm{L}}(E, V_{\mathrm{b}})\, G(E, V_{\mathrm{b}})\, \Gamma_{\mathrm{R}}(E, V_{\mathrm{b}})\, G^{\dagger}(E, V_{\mathrm{b}})\right] \tag{4.2}$$

where $G(E, V_{\mathrm{b}})$ $\left[G^{\dagger}(E, V_{\mathrm{b}})\right]$ is the retarded (advanced) Green's function and $\Gamma_{\mathrm{L,R}}$ are the coupling matrices. To compute the $I-$ V_{b} curves we have used the converged matrix density from the previous calculation to start a new calculation with a higher V_{b} Finally, the electron transmission function and the current flow through the Al chains at finite bias have been calculated using the postprocessing tool TBTrans, as included in the TranSIESTA module.

4.6.1 Transmission Coefficent

There is a clear relationship between th electronic band dispersion ,dos and the calculate transmittance because transport properties are in the ballistic regime. The transmission through a scattering region is determined by the electrode's band-structure and the energy levels for the scattering part. For instance in molecular electronics it is often useful to determine which molecular orbitals are responsible for the transmission as well as knowing their hybridization with the substrate (electrodes) While designing the gold chain two important criterions were taken into account first electrodes must be large enough to guarantee that orbitals within the unit cell only interact with one single nearest neighbor. Second the scattering region has to be large enough to avoid interaction between left electrode and right electrode.Electronic transmission function and current flowing along the wire at constant bias conditions were calculated with the help of post processing tool TBtrans of Transiesta.

The transmission function, total density of states (DOS) and projected density of states (PDOS) of gold monolayer structure at 0.1 V, 0.9 V and 1.7 V for both positive and negative bias is shown in figure 4.6 . It is clear that the transmission function, total density of states and projected density of states show a discrete band structure which varies with applied voltage.

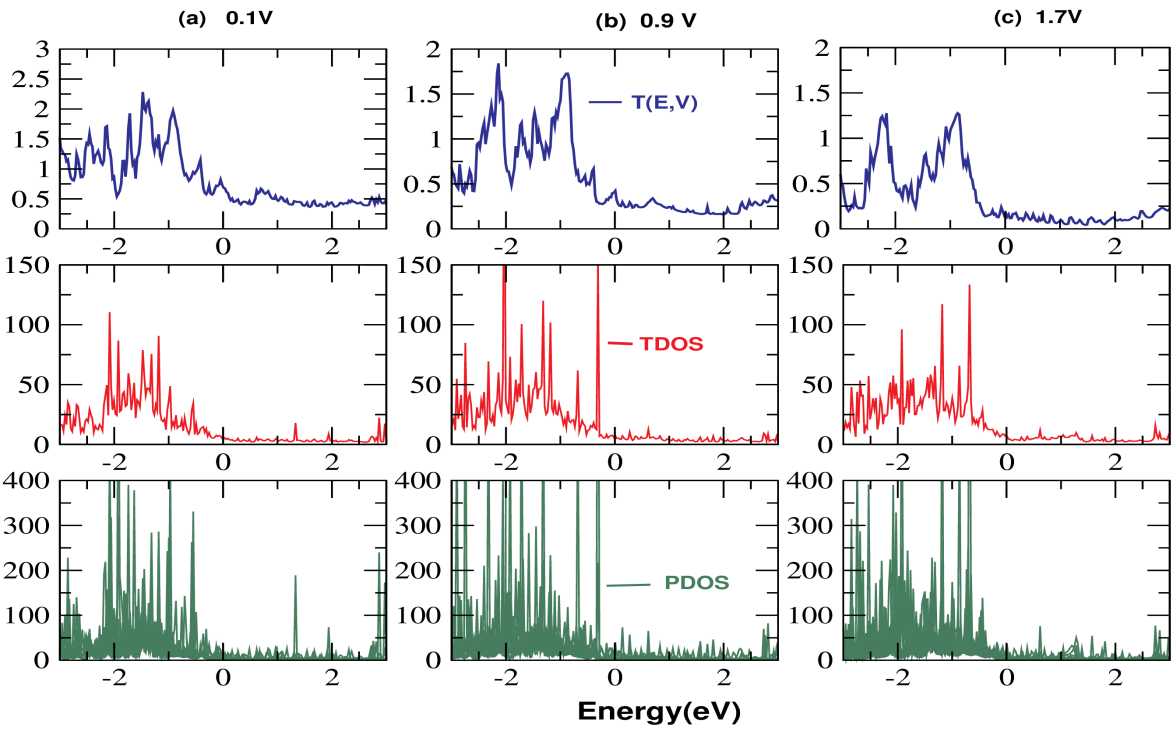

Figure 4.6: (a)Transmission function(b)total density of states (c)projected density of states for gold monolayer at positive biased voltage 0.1 V, 0.9 V and 1.7 V.

The similarity in transmission funtions suggest that there is constant current - voltage varition between both positive and negative biase.

4.6.2 Current-Voltage Characteristics

V-I characteristics were obtained by iterating the process for different bias voltage. We increased voltage in steps of 0 .1V and used the converged density matrix of the previous state as initial guess for next step [17]. Calculated projected density of state and transmission coefficient have been plotted around Fermi energy in. The overall PDOS shape of the gold monolayer decays toward high energies as is typical for a two dimensional system. However, there are strong peaks superimposed. They correspond to resonance states arising from the molecular orbitals with zero, one, two, etc. Voltage vs current graph has been shown in Figure 4.8 the $I - V_{\mathrm{b}}$ curves have been calculated with an applied bias V_{b} varied from -1.8V to $+1.8$V(Fig4.8) at the step of 0.1 V . The curve looks slightly non-symmetric in positive and negative biase. At voltages $|V_{\mathrm{b}}| < 0.4$eV, the systems behaves ohmic with a linear $I - V_{\mathrm{b}}$ characteristics.In forward

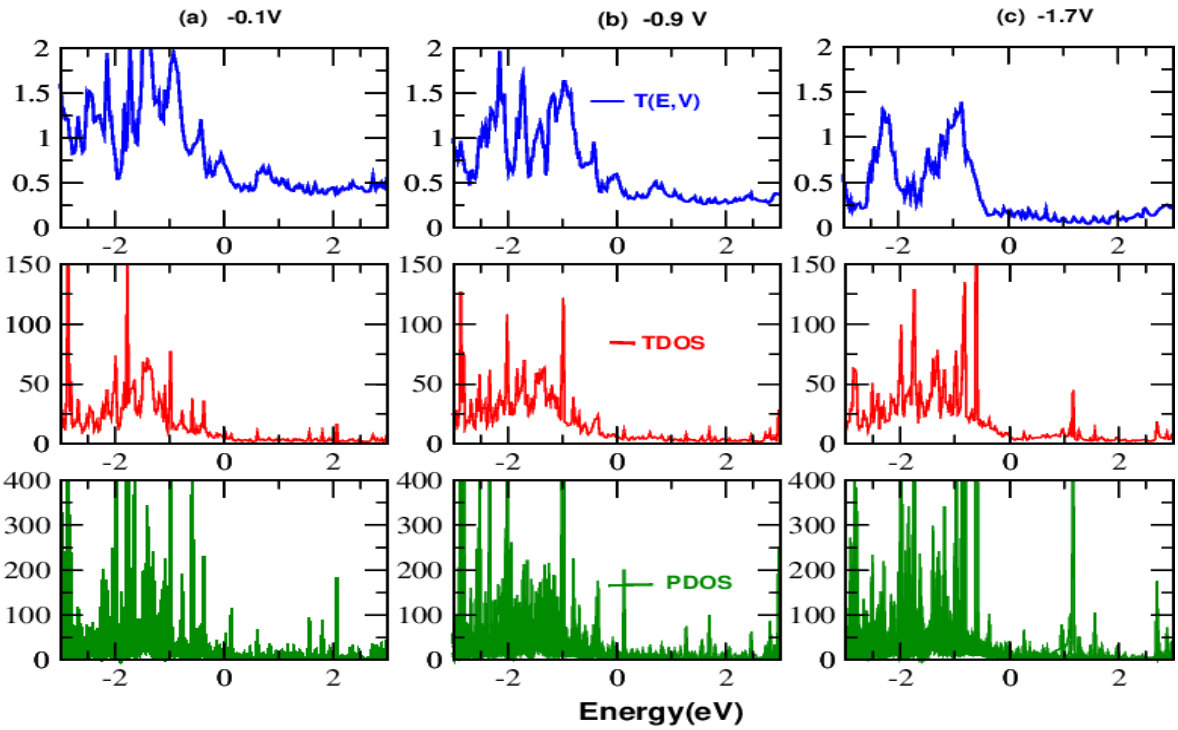

Figure 4.7: (a)Transmission function(b)total density of states (c)projected density of states for gold monolayer at negative biased voltage 0.1 V, 0.9 V and 1.7 V.

bias condition the current satuarates at between the positive bias voltage V_b 0.7V to 0.9V . Then after the 0.9V their is a hump at V_b 1.2 V on forward bias at which the current is minimum. Then after this value the current grows gradually with applied biased voltage.similary,in reverse bias condition the current saturates at between the negative bias voltage 0.85V to 0.9V with minimum current at voltage -0.1V and then after the current increases gradually on biasing. This anomalous $I-V_b$ behaviour in the gold monolayer can be ascribed as a negative differential resistance (NDR) mechanism similar to the tunnel diodes, even when this system is fully metallic. In this process the applied bias voltage induces an immediate conduction which in turn produces significant current. The current peaks at V_{peak} = 0.8V on forward bias and current peaks at V_{peak} = 0.9V with an average value of I_{peak} = 12μA. The current variation near 0 bias voltages is due to the electron quantum tunnelling. With the increase in bias voltage, the current decreases towards its minimum value, I_{valiey} = 10.66μA, at V_{valley} = 0.9V. Thus from the peak to the valley, the monolayer current falls with the increase in voltage resulting in NDR in this region. The peak-valley characteristic constitutes the most useful property of the gold monolayer.

Table 4.2: *current voltage characteristics in forward biase*

S.N	**Biase voltage(V)**	**Current (μA)**
1	0.1	2.71636
2	0.3	7.06190
3	0.5	9.81338
4	0.7	10.8250
5	0.9	11.2284
6	1.1	10.6690
7	1.3	11.1027
8	1.5	13.8051
9	1.7	16.8267

Table 4.3: *current voltage characteristics in reversed biased*

S.N	Biase voltage(V)	Current (μA)
1	-0.1	-02.755
2	-0.3	-07.425
3	-0.5	-10.720
4	-0.7	-11.683
5	-0.9	-13.271
6	-1.1	-12.754
7	-1.3	-12.721
8	-1.5	-14.917
9	-1.7	-19.504

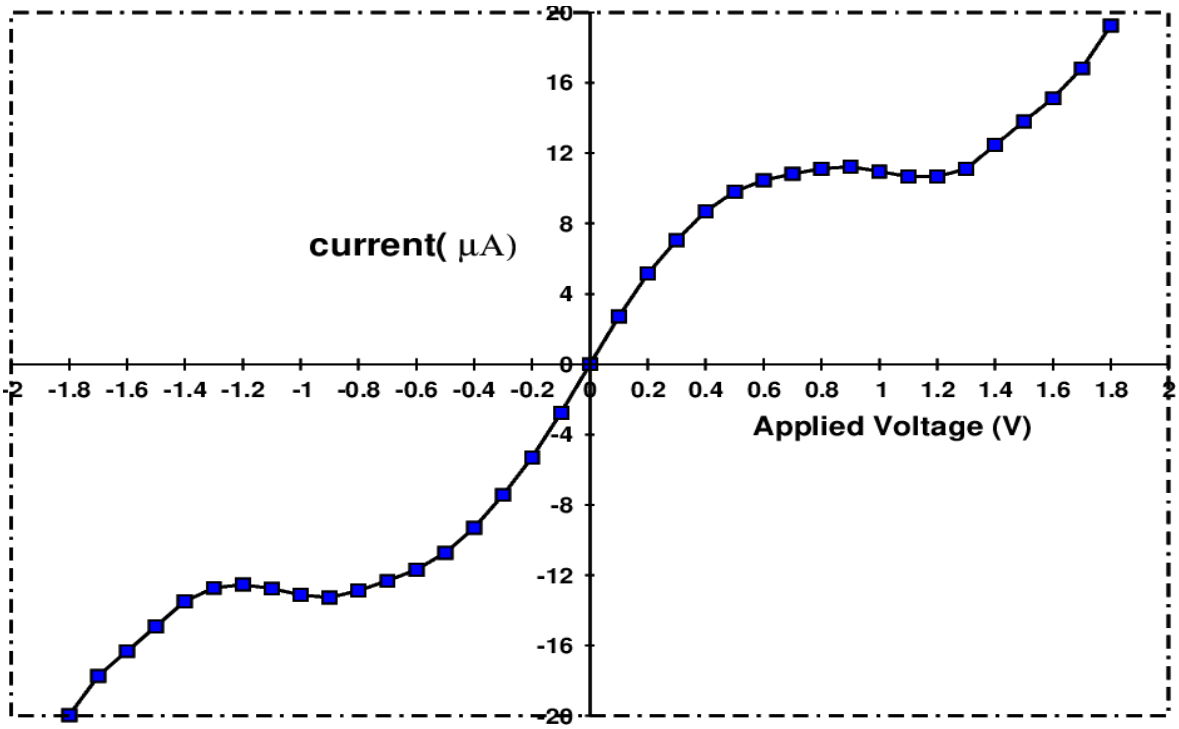

Figure 4.8: The current voltage characteristics of gold monolayer

Chapter 5

Conclusions and Future Perspectives

Our present work is electronic transport properties study of the gold monolayer(2d).All the calculation is carried out in the HPC Supercomputer facilities of Kathmandu University with SIESTA version 4.1-b4 . We performed first principle calculation based on Density Functional Theory (DFT) and Non-equilibrium Green Function (NEGF) approach within Generalized Gradient Approximation (GGA). We predict a highly stable and robust atomically thin gold monolayer with a rectangular lattice stabilized by metallic bonding.By reducing the dimension from 3D to 2D, we observed that the structural,electronic and tranport properties changes drastically. The Au monolayer has a cohesive energy of 2.98 eV per atom but the Au bulk has a cohesive energy of 3.79 eV per atom respectively.Although the cohesive energy of the 2D Au monolayer is smaller than that of the 3D bulk fcc phase,we found that monolayer has lower number of bonds per atom indicates that each bond is stronger in the sheet than in the bulk.The electronic band structure displays pure metallic nature of gold with increased number of bands crossing the fermi level.In particular we studied the transport properties of gold monolayer considering electrode-device-electrode regime.We studied the transport properties with DFT-NEGF approach using the simulation module TranSIESTA. The transport properties is studied by eliminating the current quantization effect due to the contacts. By varying the the chemical potential in the electrode region at the step 0.2 V we obtained the current -voltage characteristics of the device and also obtained the transmission coefficient. The obtained plots of PDOS,DOS and Transmission coefficent of device region is varied with different bias voltage. Impressively, the Current-voltage

plot displays a negative differential resistance (NDR) mechanism which is comparable to the Resonant Tunneling Diode(RTD) with The peak-to-valley current ratio (PVCR) 1.10.

These amazing properties of gold monolayer suggest that gold monolayer is one of the very important structure of gold for technological applications.Further,we hope that the present work can inspire the experimental fabrication of novel free standing 2D metal systems

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